

DATADOSE

PROJECT DESCRIPTION

AI-powered polypharmacy safety platform



Data Dose



Project Overview

DataDose is an enterprise-grade, AI-powered Clinical Decision Intelligence Platform that unifies graph database technology, real-time data streaming, and generative AI to deliver polypharmacy safety directly at the point of care. The platform operates across three deeply integrated layers: a live clinical application (Next.js 15 + FastAPI + Neo4j AuraDB), a complete data engineering pipeline (pandas, Aiven Kafka, Databricks PySpark, Snowflake), and an Apache Airflow orchestration layer — all containerized with Docker and unified around a shared pharmaceutical knowledge graph.

Problem Statement

Polypharmacy — patients taking five or more concurrent medications — affects over 40% of elderly patients and is a leading cause of preventable adverse drug events. Clinicians often lack real-time, context-aware tools to check interactions across a patient's complete medication list during a clinical encounter, and healthcare organizations struggle to aggregate fragmented, unvalidated EHR data for population-level analytics.

Solution

DataDose embeds a Neo4j AuraDB drug-interaction knowledge graph — spanning Drug, Disease, and Symptom nodes connected by INTERACTS_WITH, TREATS, and CAUSES_REACTION relationships — directly into the clinical workflow. Sub-second polypharmacy checks, AI-powered safe alternatives, multilingual symptom tracing, and a hybrid GraphRAG assistant are surfaced through five role-specific dashboards for Patients, Physicians, Pharmacists, Admins, and Super Admins. Behind the scenes, a five-stage validation pipeline (pandas → OpenFDA → Groq verification → Kafka → Databricks) feeds a governed Snowflake warehouse for executive Power BI reporting.

Key Features

- **Real-Time DDI Scanning:** N-degree polypharmacy scanner checking all pairwise INTERACTS_WITH relationships via Neo4j Cypher traversal, sorted by severity in milliseconds.
- **GraphRAG Clinical Assistant:** Hybrid chatbot routing clinician queries through Neo4j graph retrieval first, then LLaMA-3 synthesis, grounding responses in verified pharmaceutical data.
- **Smart Safe Alternatives:** Cypher-powered alternative drug finder given a drug to replace, a disease to treat, and symptoms to avoid — with a Groq LLM fallback when the graph returns zero results.
- **Prescription OCR Scanner:** Vision LLM (LLaMA 3.2 11B) extracts drug names from uploaded handwritten or printed prescription images.
- **Multilingual Symptom Tracer:** English and Arabic synonym expansion traces a reported adverse symptom back to the most likely suspect medication.
- **Visual Prescription Map:** React Flow canvas rendering a patient's drug list as an interactive knowledge graph of DDI, TREATS, and CAUSES_REACTION relationships.
- **Data Pipeline & Analytics:** Five-stage pandas cleaning, Aiven Kafka streaming, Databricks PySpark enrichment, and a governed Snowflake star schema feed a 4-page Power BI reporting suite.

System Architecture

The DataDose platform is built across three integrated layers — live application, data pipeline, and orchestration:

1. **Live Clinical Application:** Next.js 15 frontend and FastAPI backend, containerized with Docker Compose, delivering real-time DDI scanning and role-based dashboards.
2. **Graph Database (Neo4j AuraDB):** Stores 3,656+ Drug, Disease, and Symptom nodes, powering sub-second interaction queries via Cypher traversal.
3. **Data Processing (Databricks PySpark):** Structured Streaming consumes Kafka events, enriches each prescription against the Neo4j graph, and applies risk scoring in real time.
4. **Data Warehousing (Snowflake):** Governed star schema (staging → dimensions → facts → analytics) feeding the executive Power BI reporting suite.
5. **Orchestration (Airflow):** Local Airflow instance with its own Postgres metadata DB triggers and monitors cleaning, Kafka, and Databricks pipeline stages.
6. **Clinical Intelligence API:** 9 FastAPI REST endpoints for DDI scanning, alternatives, symptom tracing, visual mapping, OCR, and GraphRAG chat.
7. **Containerized Deployment:** Docker and Docker Compose orchestrate the frontend and backend services locally, alongside the separate Airflow container stack.

Technology Stack

- **Graph Database:** Neo4j AuraDB
- **Streaming Platform:** Apache Kafka (Aiven-managed)
- **Big Data Processing:** Databricks (PySpark)
- **Data Warehouse:** Snowflake
- **Workflow Orchestration:** Apache Airflow

- **Programming Languages:** Python, TypeScript
- **AI / LLM:** Groq (LLaMA-3, LLaMA 3.2 11B Vision)
- **Containerization:** Docker, Docker Compose
- **API Framework:** FastAPI (Python) + Next.js 15 (TypeScript)

Project Metrics

DataDose delivers measurable impact through the following:

- **Real-Time DDI Coverage:** Sub-second pairwise interaction scanning across a patient's full medication list.
- **Knowledge Graph Scale:** 3,656+ nodes spanning Drug, Disease, and Symptom entities in Neo4j AuraDB.
- **API Coverage:** 9 REST endpoints spanning scanning, alternatives, tracing, visualization, OCR, and GraphRAG chat.
- **Streaming Cadence:** Databricks Structured Streaming processes Kafka events on a 10-second micro-batch trigger, from DataDose.in through Neo4j enrichment to Snowflake staging.
- **Role-Based Access:** Five dedicated dashboards for Patient, Physician, Pharmacist, Admin, and Super Admin roles.
- **Data Pipeline Throughput:** Configurable Kafka producer rate (messages/sec, via --rate and --max flags) feeding a five-stage ETL from raw CSV through Kafka and Databricks to a governed Snowflake warehouse.

Conclusion

The DataDose Clinical Decision Intelligence Platform demonstrates how graph databases, real-time streaming, and generative AI can combine into a single, cohesive system — from a live clinical application down to a production-quality data engineering pipeline. Built with Next.js, FastAPI, Neo4j AuraDB, Kafka, Databricks, Snowflake, Power BI, Docker, and Apache Airflow, DataDose is designed for demonstration, education, and enterprise-architecture presentation purposes.